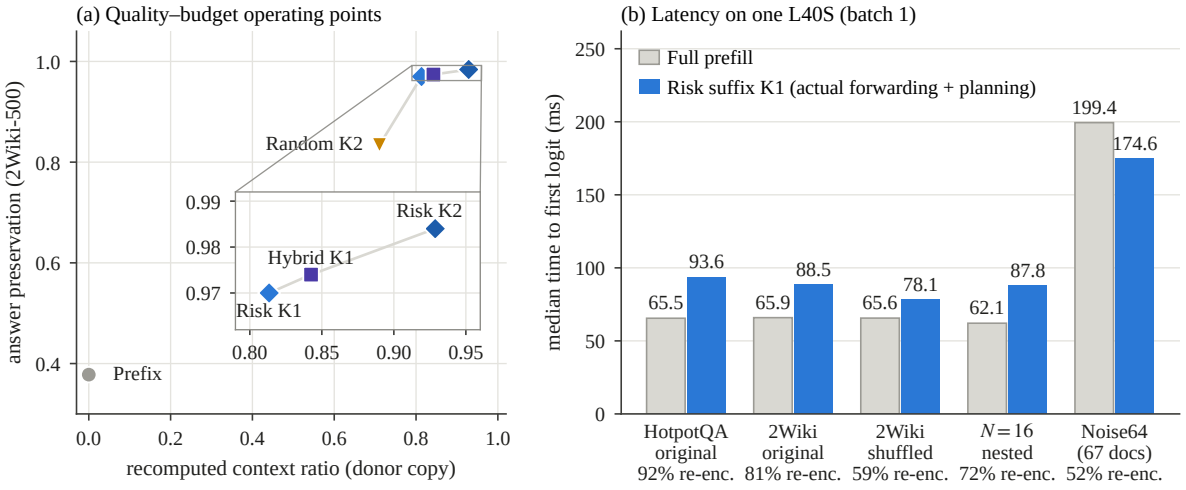
(a) Quality–budget operating points
answer preservation (2Wiki-500)
recomputed context ratio (donor copy)
1.0
0.8
0.6
0.4
0.0
0.2
0.4
0.6
0.8
1.0
Random K2
Risk K2
Hybrid K1
Risk K1
Prefix
0.99
0.98
0.97
0.80
0.85
0.90
0.95
(b) Latency on one L40S (batch 1)
median time to first logit (ms)
Full prefill
Risk suffix K1 (actual forwarding + planning)
250
200
150
100
50
0
65.5
93.6
65.9
88.5
65.6
78.1
62.1
87.8
199.4
174.6
HotpotQA original 92% re-enc.
2Wiki original 81% re-enc.
2Wiki shuffled 59% re-enc.
N = 16 nested 72% re-enc.
Noise64 (67 docs) 52% re-enc.